

Experiment 3

The Dynamic Equilibrium of Rigid Rotor

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1. Experiment Objects

- a. Gain a solid understanding of the fundamental principles and operation of dynamic equilibrium in rigid rotors.
- b. Learn how to effectively utilize a fundamental frequency detector and related instrumentation.
- c. Explore the dynamic-static method and its real-world industrial applications.
- d. Develop a clear understanding of the national standard (GB) that outlines the performance evaluation methods for dynamic equilibrium in rigid rotors and the associated requirements.

2. Experiment Contents

- a. To achieve dynamic balancing of a multi-disk rigid rotor, employ the Two-Plane-Coefficient Method (abbreviated as TPC method in subsequent sections).

3. Experiment Principles

According to the principles of dynamic-static mechanics, a rotor's inertial centrifugal force at a constant speed can be simplified into two components: a force "R" at the mass center "C" and a moment "M." When the mass center aligns with the rotation axis, and this axis is the rotor's principal axis of inertia, "R" and "M" become zero, signifying balance. If not, the rotor is imbalanced, causing strong vibrations and potential damage to the machine due to alternating reaction forces between the axis and bearings.

Balancing a rigid rotor involves reducing both the force "R" and moment "M" to zero simultaneously. This is achieved by adding two masses (m_1 and m_2) at specific distances and angles relative to a reference mark on the rotor. If these masses can counterbalance the rotor's existing forces, it's considered dynamically balanced.

The steps to apply the TPC method is as follows,

- (1) At the working speed, assess vibration parameters ($V_{10}=V_{10}\angle\psi_1$ & $V_{20}=V_{20}\angle\psi_2$) at bearings A and B. V_{10} and V_{20} are amplitudes, while ψ_1 and ψ_2 are phase angles relative to reference marks.
- (2) Choose calibration cross sections (I and II) and radii (r_1 and r_2) based on rotor structure. Add a trial mass ($Q_1=m_{t1}\angle\beta_1$) in plane I with an azimuth angle (β_1). Measure vibration parameters (V_{11} and V_{21}) at bearings A and B at the same rotating speed. α_{11} and α_{21} are influence coefficients.

$$\alpha_{11} = \frac{V_{11} - V_{10}}{Q_1}$$

$$\alpha_{21} = \frac{V_{21} - V_{20}}{Q_1}$$

(3) Remove first trial mass (m_{t1}) and add another trial mass (m_{t2}) in plane II, with azimuth angle β_2 . Measure vibration parameters (V_{12} and V_{22}) at bearings A and B). Calculate the corresponding influence coefficients.

$$\alpha_{12} = \frac{V_{12} - V_{10}}{Q_2}$$

$$\alpha_{22} = \frac{V_{22} - V_{20}}{Q_2}$$

(4) The calibration masses $p_1 = m_1 \angle \theta_1$ and $p_2 = m_2 \angle \theta_2$ and in the plane I and II. The calibration masses can be computed with the following equations,

$$\begin{cases} \alpha_{11}p_1 + \alpha_{12}p_2 = -V_{10} \\ \alpha_{21}p_1 + \alpha_{22}p_2 = -V_{20} \end{cases} \quad \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \begin{Bmatrix} p_1 \\ p_2 \end{Bmatrix} = - \begin{Bmatrix} V_{10} \\ V_{20} \end{Bmatrix}$$

(5) Apply the calibration masses to the rotor, restart it, and check if vibrations are reduced to an acceptable level. If not, repeat the procedure for further balancing.

Additionally, aside from calculated equilibrium, rotor accuracy is measured using unbalance criteria defined in GB/T9239.1-2006. It's essential to ensure that the residual unbalance U_{res} does not exceed the permissible unbalance U_{per} for effective rotor operation.

Balancing requirements vary for different rotor types and applications. Some typical balance quality standards determine the balance quality level (G) by dividing it by 2.5. For highly symmetrical rotors like the one used in this experiment, Equation $(e \cdot \Omega)$ simplifies the calculation, where m_{r1} and m_{r2} represent residual unbalance on two correction planes. For the rotor used in this experiment, the total mass of the rotor $M = 4500g$, the correction radius $R = 40mm$.

$$(e \cdot \Omega) \leq \frac{\max(m_{r1}, m_{r2}) \times 2 \times R \times \Omega}{M}$$

4. Experimental Instruments

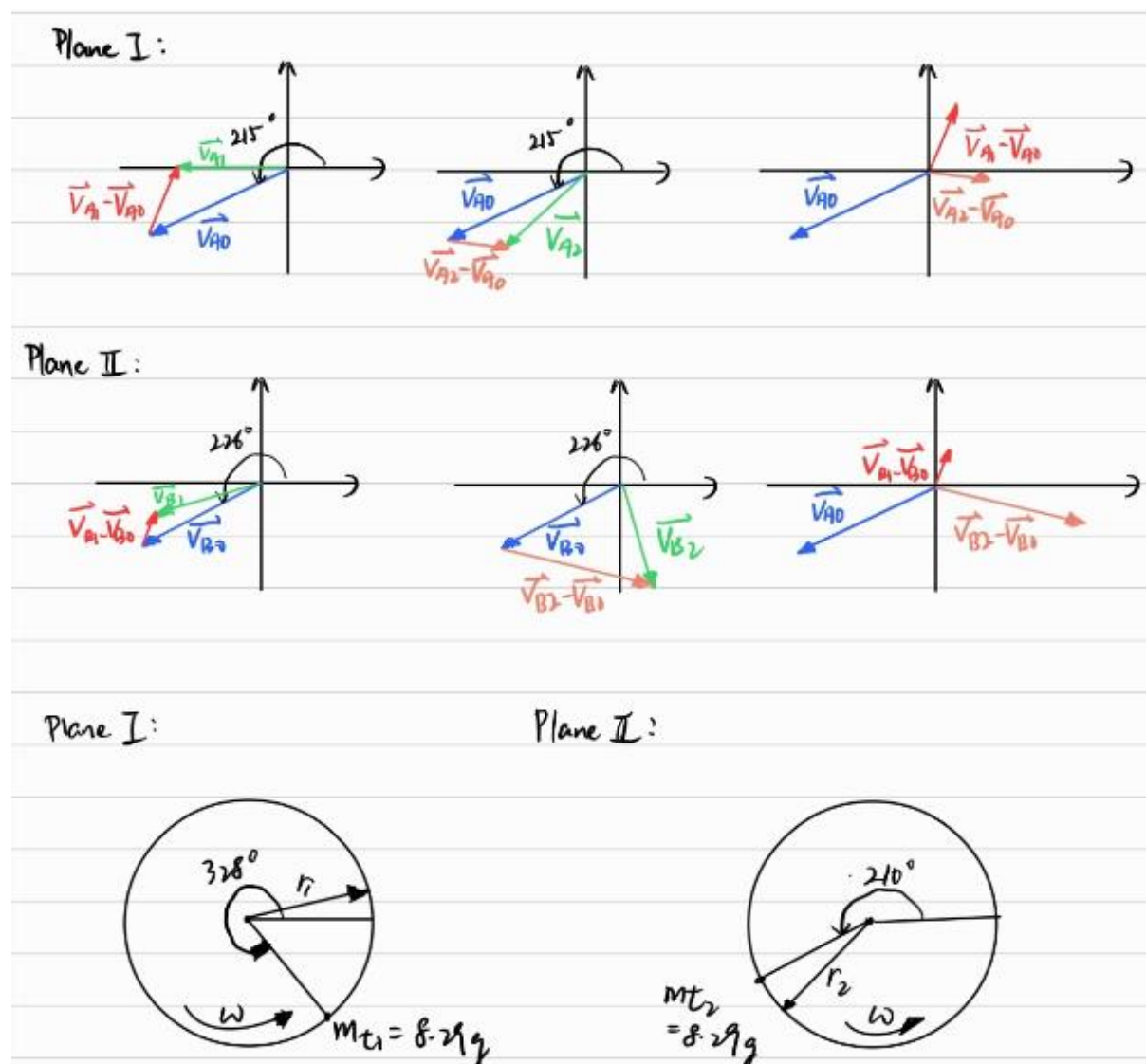
Number	Name	The main technical parameters
1	Rotor	Rotating Speed: 0~1600r/min Crucial Rotating Speed $\geq 5000r/min$
2	Speed Regulator	Speed: 600~1600r/min
3	Photoelectric Transducer	Displacement: 0.1~2000 μm
4	Eddy Current Displacement Gauge	Frequency: 0~1000Hz Displacement: 2mmVPP
5	Electric Balance	200 $\pm$ 0.01g
6	Microcomputer	A/D、D/A plug-in card is inside

5. Experiment Procedures

- (1) Launch the *Dynamic Balance for Rigid Rotors* program and click the *Virtual Connection* icon, as depicted in Fig. 5. Set the rotor speed to 1500 r/min using the speed controller.
- (2) Select the Fundamental Frequency Detection icon. Click the Start icon to initiate fundamental frequency detection. You can opt to display data from either channel A or B using the cursor. Once the readings stabilize, record the displacement amplitude and phase angles for bearings A and B caused by the initial imbalance.
- (3) Halt the rotor. Place a trial mass (m_{t1}) on Plane I at any position. Record the weight of the trial mass (preferably between 8-10g) and the phase angle of its placement.
- (4) Restart the rotor. Restore the rotor speed to its previous value. Measure the amplitude and phase angles of the vibration displacement at the two bearings influenced by the added mass on Plane I.
- (5) Halt the rotor. Remove m_{t1} . Introduce a trial mass (m_{t2}) at an arbitrary location on Plane II. Record the weight of the trial mass and the phase angle of its placement.
- (6) Reset the rotor's speed to its previous value. Measure the amplitude and phase angles of the vibration displacement at the two bearings with the mass added on Plane II. If the vibration substantially surpasses the initial condition, adjust the phase of the trial mass.
- (7) Stop the rotor. Remove the trial masses. Click the Dynamic Balance Calculation icon. Enter the data following the software instructions. Record the computed calibration masses.
- (8) Based on the calibration masses and their calibration angles, introduce the necessary masses on reference Planes I and II. Then reset the rotor's speed to its previous value. Measure the amplitude and phase angles of the remaining vibration.
- (9) Halt the rotor. Return to the Dynamic Balance Calculation interface. Calculate the equilibrium percentage, residual unbalance, and the achieved balance accuracy level.
- (10) Calculate the "synthetic correction quality" for both calibration planes.
- (11) Utilize "Synthetic Calibration Quality" in place of "Calibration Quality" and repeat steps (8) and (9).
- (12) Power down the rotor and the instruments. Remove the calibration masses.

6.Experiment Data

Stabilized rotating speed: 1500 r/min		Bearing A/ Plane I		Bearing B/ Plane II	
		Amplitude / Mass	Phase angle	Amplitude / Mass	Phase angle
Dynamic Equilibrium of Rotor	Initial Vibration Quantities V_{A0}, V_{B0}	$6.6\mu\text{m}$	215°	$5.9\mu\text{m}$	226°
	Trial Mass Q_1 (Plane I)	8.29g	328°	-	-
	Amplitude and Phase Angle of Vibration Displacement V_{A1}, V_{B1}	$6.6\mu\text{m}$	180°	$5.5\mu\text{m}$	216°
	Trial Mass Q_2 (Plane II)	-	-	8.29g	210°
	Amplitude and Phase Angle of Vibration Displacement V_{A2}, V_{B2}	$6.2\mu\text{m}$	220°	$5.9\mu\text{m}$	278°
	Calibration Masses P_1, P_2	12.73g	254°	7.27g	278°
	Residual Vibration	$0.13\mu\text{m}$	28°	$0.15\mu\text{m}$	235°
	Equilibrium Percentage	98.03%	-	97.46%	-
	Residual Imbalance Mass	0.32g	72°	0.30g	278°
Equilibrium Iteration	Composite Calibration Mass	12.41g	254°	7.27g	278°
	Residual Vibration after Iteration	$0.19\mu\text{m}$	322°	$0.056\mu\text{m}$	236°
	Equilibrium Percentage after Iteration	97.12%	-	99.05%	-
	Residual Imbalance Mass after Iteration	0.41g	5°	0.12g	241°



(3) Known conditions: $m_{r1} = 0.41g$, $m_{r2} = 0.12g$, $\omega = 1500 \text{ rad/s}$, $M = 4500g$

(e.g. ω) value: $\frac{U \times \omega}{M} \leq \frac{\max(m_{r1}, m_{r2}) \times 2 \times R \times \omega}{m} = \frac{0.32 \times 2 \times 40 \times 1500 \times \frac{2\pi}{60}}{4500}$

$= 1.1449$

$\therefore$ Balance quality level G2.5

(4) i) Dynamic Balancing Facilities (DBF) are typically equipped with high-speed balancing machines ("low speed" – less than 2000 rpm). The level of accuracy of machines in DBF is also higher than instruments in lab. High speed balancing is required when additional vibration modes (critical speeds) or radial growth of components are encountered at operational speed. Machines in DBF can go as fast as 15500 rpm. Instruments in lab can go faster than 1500 rpm but it's still no where close to 15500 rpm. The focus of DBF is on achieving high levels of accuracy and efficiency for mass production whereas instruments in lab are designated for academic or experimental work.

ii) The scale of DBF is way bigger than instruments in lab. DBF are capable of balancing a wide range of industrial machinery and components, including large rotors, driveshafts, and turbines. They can handle heavy loads and may use automated systems to achieve high efficiency and production rates.

(5)

刚性转子动平衡							
姓名		班级		学号		组号:	
杨礼谦		机械24		2022080054			
						2023 年 11 月 01 日	
平衡转速: 1500 rpm		A 轴承/I 平面			B 轴承/II 平面		
		幅值/质量		相位		幅值/质量	相位
实验 工况	转子 动平衡	初始振动量		6.6 微米	215 deg	5.9 微米	226 deg
				8.29 克	328 deg		
				6.6 微米	180 deg	5.5 微米	216 deg
						8.29 克	210 deg
				6.2 微米	220 deg	5.9 微米	278 deg
				12.73 克	254 deg	6.97 克	278 deg
				0.13 微米	28 deg	0.15 微米	235 deg
				98.03 %			%
			0.32 克	72 deg	0.30 克	278 deg	
	平衡 迭代	合成校正质量		12.41 克	254 deg	7.27 克	278 deg
		0.19 微米	322 deg	0.056 微米	236 deg		
		97.12 %			99.05 %		
		0.41 克	5 deg	0.12 克	241 deg		
平衡品质级别		$\frac{0.41 \times 2 \times 40 \times 500}{4500} = 1.1449$ G2.5					